DSRKD: Joint Despecking and Super-Resolution of SAR Images via Knowledge Distillation

Ziyuan Liu, Shaoping Wang, Ying Li, Yuantao Gu, *Senior Member, IEEE*, and Quan Yu, *Fellow, IEEE*

***Abstract*— Deep learning has achieved success in optical image super-resolution (SR). However, few methods have been proposed for synthetic aperture radar (SAR) images. This is because SAR images inherently suffer from severe speckle noise, and their resolution is typically much lower compared with optical images, making SR more challenging. To simultaneously denoise and preserve texture details in SAR image SR tasks, we propose a joint despeckling and SR network via knowledge distillation (DSRKD). By implementing feature distillation (FD) in the encoding stage, we enable the student network to obtain noise-free latent variables, thereby restoring clean SR images, and simultaneously enhancing effective supervision for the student using target distillation (TD). This distillation strategy does not incur additional computational costs during inference. Experimental results demonstrate that our algorithm effectively suppresses speckle noise while preserving texture details of images on both synthetic and SAR datasets. Compared with other state-of-the-art algorithms, the proposed method achieves superior performance on both visual results and quantitative metrics, such as PSNR, SSIM, and other nonreference indicators under various SR scales and speckle intensities.**

***Index Terms*— Image despeckling, image super-resolution (SR), knowledge distillation (KD), synthetic aperture radar (SAR).**

I. Introduction

A SYNTHETIC aperture radar (SAR) is an active microwave sensor capable of all-weather and continuous ground observation with various applications. High-resolution (HR) SAR images are urgently needed for tasks, such as target identification, object tracking, and image classification. However, SAR image super-resolution (SR) has always been a challenging problem due to the corruption of inherent speckle noise in SAR images.

Single-image SR (SISR) aims to recover HR images from low-resolution (LR) images. Most SR methods typically utilize an ideal bicubic downsampling kernel to generate LR images [1], [2], [3], [4], [5], [6]. However, this simple degradation does not apply in SAR scenarios. When applying these SR methods directly to SAR images, the speckle noise is also amplified, leading to more severe contamination of the images.

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Ziyuan Liu and Yuantao Gu are with the Department of Electronic Engineering, Beijing National Research Center for Information Science and Technology, Tsinghua University, Beijing 100084, China (e-mail: liuziyua22@mails.tsinghua.edu.cn; gyt@tsinghua.edu.cn).

Shaoping Wang is with the Tsinghua Shenzhen International Graduate School, Shenzhen 518055, China (e-mail: wsp22@mails.tsinghua.edu.cn).

Ying Li and Quan Yu are with Peng Cheng Laboratory, Shenzhen 518066, China (e-mail: liy02@pcl.ac.cn; yuq@pcl.ac.cn).

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A natural and direct solution is to achieve joint despeckling and SR of SAR images by cascading denoising and SR algorithms, since denoising algorithms for SAR images have been widely researched [7], [8], [9], [10], [11], [12]. However, the simple combination of separate models for this joint task may not produce optimal results. On the one hand, sequential methods introduce additional time overhead due to intermediate results, resulting in lower efficiency. Lower efficiency limits practical applications, especially in mobile SAR devices. On the other hand, sequential solutions are susceptible to error accumulation and information loss in each model. For an LR SAR image, performing SR first amplifies noise, making subsequent denoising more difficult; performing denoising first directly on the LR image may excessively smooth image details, resulting in the subsequent SR results being blurrier. Therefore, we need a joint method that can perform despeckling and SR simultaneously. In the field of optical images, some works on denoising and SR of images contaminated by additive Gaussian noise or block artifacts caused by JPEG compression have been proposed [3], [13]. However, research on denoising and SR of SAR images contaminated by speckle noise is few. A recent work attempted to solve this task by cascading denoising and reconstruction networks [14], but this simple cascading approach does not yield optimal results.

In this article, we propose a joint despeckling and SR method via knowledge distillation (DSRKD) for SAR images. The overall pipeline of DSRKD is depicted in Fig. 1. Specifically, we design an SR network with an encoder–upscaling–decoder architecture. We first input clean paired HR and LR images to the teacher model DSRKD_T, enabling the teacher to fully learn the distribution of the original images. When training the student model DSRKD_S with speckled LR images, we use feature distillation (FD) in the encoder phase to enable the student to learn the distribution of noise-free latent variables from the teacher, thus decoding clean SR images. In addition, we introduce target distillation (TD), enabling the teacher's outputs and training data to jointly supervise the student for better performance. Extensive experiments on both synthetic and SAR data demonstrate the effectiveness of the proposed DSRKD. It effectively removes noise and preserves image details when super-resolving speckled images. Our main contributions are as follows.

1) We propose the DSRKD network, with an encoder–upscaling–decoder architecture that allows for SR at various resolution scales. The encoder and decoder can also be replaced with various more complex modules, and

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various distillation methods can be easily added, making it a simple and efficient baseline for despeckling SR.
2) We innovatively combine the SR of speckled images with denoising using knowledge distillation (KD), significantly improving the performance of the model without introducing any additional computational overhead during inference.
3) Extensive experiments on synthetic and real-world SAR data show that the proposed DSRKD achieves excellent performance on various combinations of datasets, resolutions, and speckle intensities.

II. Related Works

A. SAR Image Super-Resolution

The azimuth resolution of SAR is approximately half of the antenna length. Reducing the antenna length can enhance the azimuth resolution. However, this also results in a decrease in antenna gain and detection range. The range resolution of SAR is inversely proportional to the signal bandwidth. Nonetheless, the bandwidth is constrained by the chirp rate and pulselength, limiting the extent to which range resolution can be improved. As a result, the resolution of SAR images is often significantly lower than that of optical images. Consequently, achieving SR reconstruction of SAR images is even more challenging. Traditional SAR image SR algorithms mostly rely on parametric spectral estimation or compressed sensing (CS) theory frameworks [15], [16]. In recent years, deep-learning-based SR has been successfully applied to optical images, including remote sensing images [17], [18]. However, due to the corruption of multiplicative noise known as speckle in SAR images, there are limited deep learning methods for SAR image SR, and the reconstruction results are often suboptimal.

Wei et al. [19] combined alternating direction method of multipliers (ADMM) with neural networks to achieve fast SR of SAR images. Luo et al. [20] proposed an SAR image SR method based on the traditional fast SR convolutional neural network (FSRCNN), introducing the structural similarity index (SSIM) into the loss function, and achieved good results in both subjective visual and objective evaluation metrics. Wang et al. [21] utilized the generative adversarial network for SAR image SR (referred to as SRGAN). Compared with traditional SR algorithms, SRGAN achieved significant breakthroughs in reconstruction accuracy and computational efficiency. Addressing the issue of low spatial resolution in polarimetric SAR (PolSAR) images, Shen et al. [22] proposed a PolSAR SR framework using residual convolutional neural networks to enhance spatial resolution. Yanshan et al. [23] proposed a novel optical-guided SR network (OGSRN) for SAR images with large scaling factors. Guided by the corresponding optical images of SAR images, OGSRN achieved excellent performance in both quantitative evaluation metrics and visual quality.

However, most previous works did not consider the influence of speckle noise. When SAR images are super-resolved, the speckles are also enlarged, resulting in worse contamination. In contrast to these methods, our approach can simultaneously perform joint denoising and SR to obtain noise-free SR SAR images.

B. SAR Image Despeckling

Due to the coherent summation of many elementary echoes within the radar resolution cell, SAR images are often severely contaminated by speckle noise. Noise-free SAR images are not available in reality. Hence, SAR image denoising has always been a challenging problem. Parrilli et al. [7] took into account the peculiarities of SAR images and extended the block-matching 3-D (BM3-D) [24] algorithm to SAR image denoising. Based on the nonlocal-means [25] algorithm, Deledalle et al. [8] proposed a general method called NL-SAR for denoising various types of SAR images.

Deep learning is currently the most popular method for solving computer vision problems. Therefore, it has also been widely applied to SAR image despeckling in recent years and has achieved good results. Since there is no ground truth for SAR images, deep-learning-based denoising methods can be divided into two types: self-supervised learning and supervised learning using simulated data. Molini et al. [9] proposed a self-supervised Bayesian despeckling method based on the theory of blind-spot denoising networks [26]. Dalsasso et al. [10] introduced a semisupervised despeckling network called SAR2SAR, which is based on the Noise2Noise model [27]. They also proposed MERLIN, which is a self-supervised strategy for denoising single-look complex (SLC) SAR images by separating the real and imaginary parts, and can be used to train various deep networks [28].

For supervised SAR image despeckling, the networks are usually trained on synthetic datasets or multitemporal SAR data. Chierchia et al. [11] were the first to propose the end-to-end CNN-based SAR image denoising network. Liu et al. [12] proposed the multiscale residual dense dual attention network (MRDDANet), which effectively removed speckles while preserving the texture details of the image. Vitale et al. [29] proposed MONet for SAR despeckling, effectively learning the distribution of speckles by designing a multiobjective cost function. They then achieved better results in denoising SAR images through more refined modeling of SAR scenarios and new multiobjective loss function design [30].

C. Multitask Joint Image Processing

In both SAR and optical image processing pipelines, a series of operations are typically included. Naively cascading various image processing operations can lead to an accumulation of errors from each method. Therefore, it is necessary to study joint solutions for various task combinations.

Recently, in optical image processing tasks, the joint execution of SR and denoising has been proposed [3], [13], where the noise studied usually includes common Gaussian noise or block artifacts caused by JPEG compression. Lei et al. [31] proposed a solution to simultaneously address the SR and segmentation tasks for remote sensing images. Due to the severe speckle noise contamination in SAR images, some methods for joint processing of SAR image tasks have also been proposed. Liu et al. [32], [33] achieved joint despeckling

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and compression of SAR images using a two-stage training approach or KD. Gu et al. [14] proposed a joint denoising and SR algorithm for SAR images based on the GAN model, maintaining SAR image details under high levels of noise suppression. However, their method, which simply concatenates the denoising network and the reconstruction network, may not obtain optimal results. Different from their work, we utilize KD to train a complete SR network, which can achieve better despeckling performance.

D. Knowledge Distillation

KD is a concept proposed by Hinton et al. [34], aimed at transferring the representational capability of a teacher network to a student network. KD was initially introduced for image classification tasks, following a teacher–student framework, which transfers knowledge from a powerful teacher network to a simple network, achieving model compression with slight performance loss. Gou et al. [35] categorized distilled knowledge into three types: target knowledge, which refers to the soft targets output by the teacher model, commonly used for classification tasks; feature knowledge, which is learned from the intermediate outputs of the teacher model; and relationship knowledge, which denotes learning the relationships between different layers or data samples. Recently, KD has also been introduced into optical image SR tasks. Gao et al. [36] applied basic FD directly to image SR and compared the impact of different statistical quantities on the performance of the distilled models. He et al. [37] used the feature correlation idea for KD, designing an innovative feature-affinity-based KD (FAKD) loss, which improves the performance of various SR networks. Lee et al. [38] further introduced variational information distillation (VID), maximizing the mutual information between the intermediate feature maps of the teacher and student SR networks, enabling the student to effectively learn the knowledge of the teacher model. In SAR scenarios, some KD-based detection and classification tasks have been proposed. Zhou et al. [39] proposed an ultralightweight SAR ship detector based on distillation technology, reducing the model's parameter quantity while maintaining high detection accuracy. Xu et al. [40] improved the performance of land cover classification by introducing multimodal KD. However, the KD technology has not yet caught much attention in SAR image SR.

Most previous works utilized KD solely for model compression. In contrast, we developed DSRKD to simultaneously perform student model compression and learning of despeckling capabilities.

III. Proposed Method

In this section, we will comprehensively present the proposed method. We first introduce the architecture of the DSRKD network in Section III-A, followed by a detailed description of our distillation strategy in Section III-B. Finally, we introduce the overall loss function of training the teacher and student models in Section III-C.

A. Network Architecture

To facilitate KD, the student and teacher networks share the same architecture, while the number of channels in each layer of the student is half that of the teacher. As illustrated in Fig. 2, the proposed despeckling–SR network (DSRKD) is composed of three components: an encoder, an upscaling module, and a decoder. For an original image X, the encoder first transforms it into the latent feature space. The upscaling module is essential for SR at arbitrary resolutions, while the decoder maps the super-resolved latent representations back to the image domain. We utilize a series of MultiResBlocks (MRBs) to extract image features from coarse to fine. Attention modules (AMs) are plugged into the network to enhance the ability to highlight important features. The details of MRB and AM are shown in Fig. 3. To prevent the loss of edge and texture information, we fuse the outputs of the shallow layers into subsequent network layers. Consequently, a series of latent features are obtained during the encoding stage, as formulated below

$$\begin{aligned} Z_1 &= E_1(\mathrm{Conv}(X)) \\ Z_2 &= E_2(Z_1 + \mathrm{Conv}(X)) \\ Z_3 &= E_3(Z_2 + \mathrm{Conv}(X)) \end{aligned} \tag{1}$$

where X is the image fed to the network, E_1 and E_2 consist of two MRBs cascaded with an AM, and E_3 is composed of two MRBs. Subsequently, the upscaling module utilizes the multilevel features for subpixel operation and fuse them to obtain the coarse SR features

$$Z_{\mathrm{SR}} = S(Z_1) + S(Z_2) + S(Z_3) \tag{2}$$

where S denotes the function of subpixel convolution. Finally, the SR features Z_{SR} are fed into the decoder to reconstruct the SR image $\hat{X}$.

B. Distillation Strategy

For an encoder-based and decoder-based network, regardless of the input variations, if we can obtain the same latent variables using whatever methods, the decoded outputs will be identical. Specifically, for an SR network, increased proximity in latent variations can lead to improved proximity in the reconstructed images. In light of this, we design the DSRKD network with an encoder–upscaling–decoder structure. The KD strategy is employed during the encoding process to remove speckles, ensuring the acquisition of noise-free latent variables and subsequently reconstructing clean SR images. Fig. 1 depicts the synthetic process of training data and the diagram of our distillation strategy. The training process of DSRKD consists of two stages.

1) Teacher Training: Since we need noise-free latent variables as the "knowledge" for despeckling guidance, the teacher DSRKD$_T$ is trained as a pure SR model using clean data. However, the process of downsampling HR images to obtain LR images leads to significant information loss. To enable DSRKD$_T$ to fully learn the distribution of HR images for better guidance of the student, we perform the Pixel-Unshuffle operation on the HR image $X_{\mathrm{HR}} \in R^{1\times rH\times rW}$ to obtain

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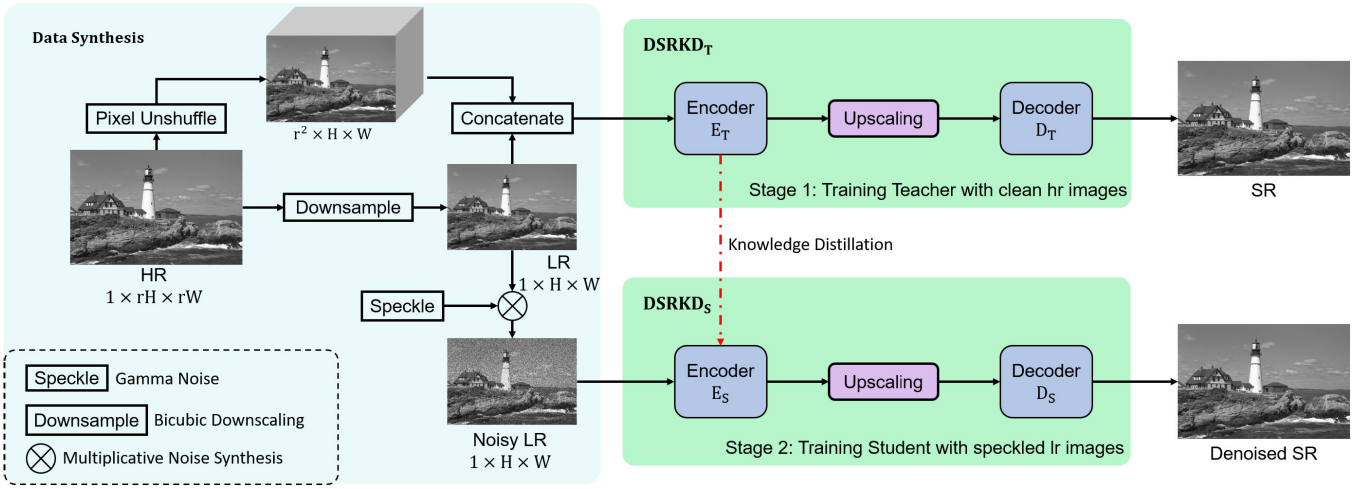


Fig. 1. Overview of the process of training data synthesis and KD mechanism. Clean LR and HR images are fed into the teacher DSRKD$_T$, while the corresponding speckled LR images are fed into the student DSRKD$_S$.

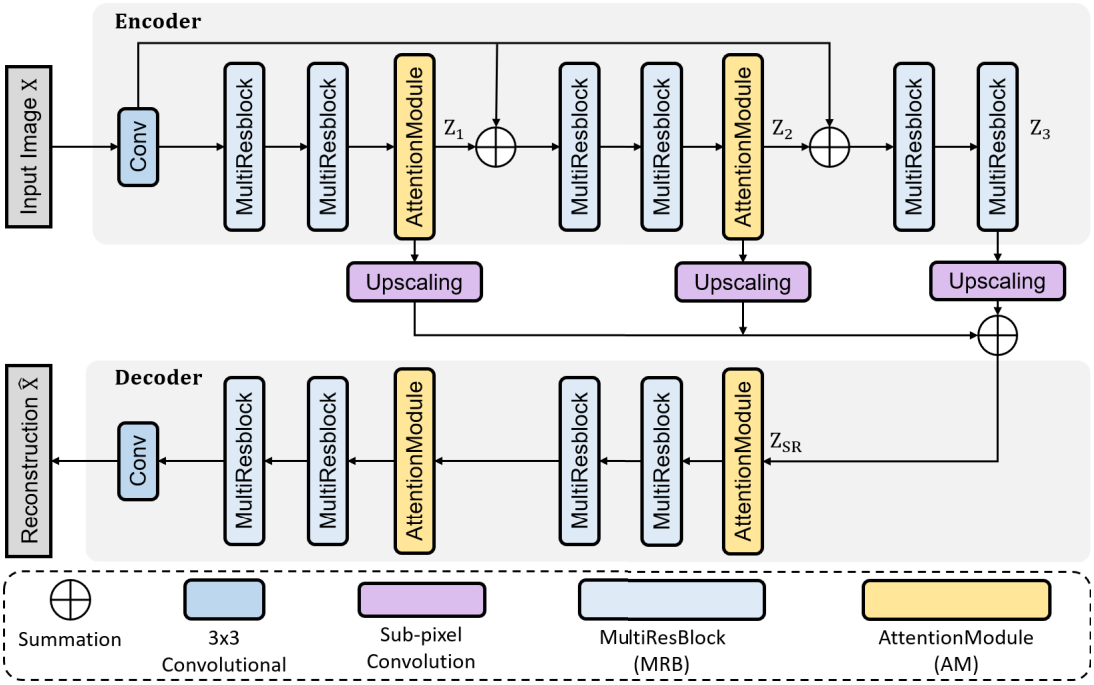


Fig. 2. Overall architecture of the proposed joint DSRKD. The teacher and student networks share the same architecture, with the number of channels in each layer of the student network being half that of the teacher network.

$X'_{\mathrm{HR}} \in R^{r^2 \times H \times W}$, which is then concatenated with LR images $X_{\mathrm{LR}} \in R^{1 \times H \times W}$ as the input $X \in R^{(r^2+1) \times H \times W}$ for the teacher network. Therefore, DSRKD$_T$ can be considered a reconstruction network strictly speaking.

2) Student Training: During the training of the student DSRKD$_S$, speckled LR images X^n_{LR} are fed into the student network, while the corresponding clean LR images X_{LR} and HR images X_{HR} are fed into the teacher DSRKD$_T$. Then, the distillation strategy is employed. We select two distillation components: FD and TD.

The purpose of FD is to make the latent representations of the student and teacher as similar as possible, enabling the student to reconstruct noise-free SR images. Therefore, FD is exclusively performed during the encoding phase. To validate the effectiveness of our model with different distillation methods, we select two distillation loss functions: the L_1 norm and feature-affinity KD (FAKD) [37].

When using the L_1 norm directly as the FD loss, the number of channels in the distilled feature maps of the teacher network and the student network must be the same. In this scenario, the FD loss can be formulated as follows:

$$L_{\mathrm{FD}} = \frac{1}{|Z|} \sum_{l=1}^{N} \left|\left| Z_l^S - Z_l^T \right|\right|_1 \tag{3}$$

where Z_l^S and Z_l^T are the lth feature layers of DSRKD$_S$ and DSRKD$_T$, $|Z|$ denotes the number of elements in the feature layers, and N is the number of layers we choose to distill.

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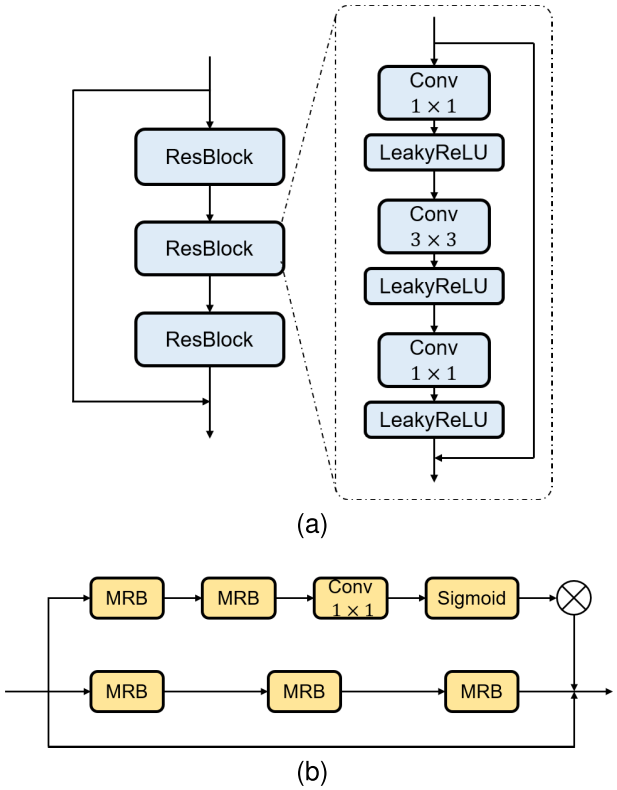


Fig. 3. Details of MRB and AM. (a) MRB. (b) AM.

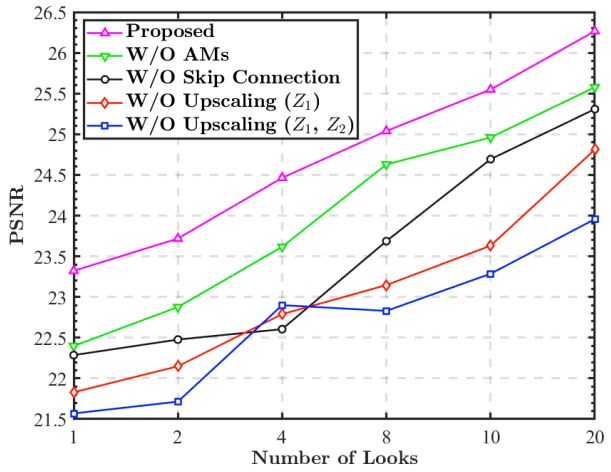


Fig. 4. Effects of different DSRKD network modules on the performance. The PSNR values are measured on the KodaK24 dataset across six different looks.

To obtain a more powerful teacher model, we also need to consider the scenario where the number of channels in the teacher model's network layers exceeds that of the student model. For this purpose, we introduce the FAKD algorithm. Specifically, given a batch of feature maps $Z \in R^{N\times C\times W\times H}$, we first reshape it into a 3-D tensor $Z \in R^{N\times C\times WH}$ and normalize it to $\tilde{Z}_{[i,:,j]} = Z_{[i,:,j]}/||Z_{[i,:,j]}||_2^2$. Then, we obtain the spatial affinity matrix by inner product $A = \tilde{Z}^T \cdot \tilde{Z}$. In this case, the FD loss can be formulated as follows:

$$L_{\mathrm{FD}} = \frac{1}{|A|}\sum_{l=1}^{N}\left|\left|A_l^S - A_l^T\right|\right|_1. \tag{4}$$

Similarly, A_l^S and A_l^T are the affinity matrix of DSRKD$_S$ and DSRKD$_T$ extracted from the feature maps of the lth layer, and $|A|$ denotes the number of elements in the affinity matrix. In this study, as illustrated in Fig. 2, the images decoded by the model are completely determined by Z_{SR}, which is the combination of the upsampled Z_1, Z_2, and Z_3 produced by the encoder. Consequently, we perform FD exclusively on the feature layers $Z_1 \sim Z_3$, enabling the student model to learn noise-free features for subsequent upsampling and reconstruction.

Besides, we observe that supervision from the teacher's reconstructed images enhances the performance of the student. Therefore, to enable DSRKD$_S$ to receive the distribution information from both real data and DSRKD$_T$ during training, we also introduce the TD loss to minimize the L_1 distance between the reconstructed images of DSRKD$_S$ and DSRKD$_T$

$$L_{\mathrm{TD}} = \left|\left|f_{\mathrm{DSRKD}_S}\left(X_{\mathrm{LR}}^n\right) - f_{\mathrm{DSRKD}_T}(X_{\mathrm{LR}}, X_{\mathrm{HR}})\right|\right|_1 \tag{5}$$

where f_{DSRKD_S} and f_{DSRKD_T} are the functions of DSRKD$_S$ and DSRKD$_T$.

C. Loss Function

When training the teacher DSRKD$_T$, we only require it to perform a reconstruction task, so L_1 norm is introduced as the reconstruction loss, since its efficiency has been widely recognized in such tasks [30]. The loss function for training DSRKD$_T$ can be formulated as follows:

$$L_T = ||f_{\mathrm{DSRKD}_T}(X_{\mathrm{LR}}, X_{\mathrm{HR}}) - X_{\mathrm{HR}}||_1. \tag{6}$$

When training the student DSRKD$_S$, we not only require real data for supervision but also need distillation guidance from the teacher. We perform FD in each AM layer within the encoder and simultaneously introduce TD loss and reconstruction loss. Therefore, the overall loss function for training DSRKD$_S$ can be formulated as follows:

$$\begin{aligned} L &= \left|\left|f_{\mathrm{DSRKD}_S}\left(X_{\mathrm{LR}}^n\right) - X_{\mathrm{HR}}\right|\right|_1 \\ L_S &= L + \lambda_1 L_{\mathrm{TD}} + \lambda_2 L_{\mathrm{FD}} \end{aligned} \tag{7}$$

where λ_1 and λ_2 govern the trade-off among different aspects of the loss.

TABLE I

EFFECTS OF DIFFERENT DISTILLATION COMPONENTS ON THE PERFORMANCE OF THE STUDENT DSRKD$_S$ (BEST AND SECOND-BEST PERFORMANCES ARE IN RED AND BLUE, RESPECTIVELY)

Component		PSNR				
FD	TD	Set5	Set14	BSD100	Urban100	KodaK24
✗	✗	26.402	24.435	23.887	23.367	25.079
L_1	✗	26.874	24.752	24.013	23.799	25.464
FAKD	✗	26.975	24.767	24.104	23.811	25.499
✗	L_1	26.855	24.747	24.108	23.699	25.395
L_1	L_1	26.913	24.799	24.110	23.919	25.487
FAKD	L_1	27.052	24.894	24.220	24.334	25.549

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TABLE II
COMPARISON OF MODEL PARAMETER COUNTS AND RUNTIME FOR DIFFERENT METHODS AT ×2 UPSCALING

Methods	SRCNN	FSRCNN	CFSRCNN	MemNet	RCAN	RRDBNet	NFGAN	DSRKD$_S$
Parameters (K)	56	12	1200	1774	15810	17097	3092	1773
Time (ms)	210.6	15.2	160.6	397.9	365.7	407.9	750.2	184.8

TABLE III
COMPARISON OF AVERAGE PSNR/SSIM PERFORMANCES OF VARIOUS DESPECKLING SR METHODS FOR ×2, ×3, AND ×4 UPSCALING WITH SPECKLES OF TEN LOOKS (BEST AND SECOND-BEST PERFORMANCES ARE IN RED AND BLUE, RESPECTIVELY)

Datasets	Scale	SRCNN	FSRCNN	CFSRCNN	MemNet	RCAN	RRDBNet	NFGAN	Proposed
Set5	×2	13.43 / 0.476	26.09 / 0.758	27.04 / 0.797	24.47 / 0.710	26.23 / 0.780	27.01 / 0.806	24.17 / 0.688	27.05 / 0.798
	×3	13.53 / 0.466	24.56 / 0.716	25.13 / 0.739	21.66 / 0.605	24.39 / 0.723	25.10 / 0.751	22.89 / 0.588	26.45 / 0.769
	×4	13.21 / 0.455	23.34 / 0.673	23.81 / 0.686	20.70 / 0.559	23.03 / 0.676	23.74 / 0.707	21.64 / 0.574	25.80 / 0.740
Set14	×2	14.64 / 0.449	24.56 / 0.650	24.80 / 0.679	21.52 / 0.534	23.85 / 0.657	25.06 / 0.694	22.58 / 0.570	24.89 / 0.681
	×3	14.76 / 0.443	23.28 / 0.601	23.28 / 0.611	20.48 / 0.500	22.27 / 0.591	23.33 / 0.623	21.63 / 0.488	24.14 / 0.633
	×4	14.46 / 0.433	22.33 / 0.570	22.30 / 0.568	20.32 / 0.504	21.34 / 0.552	22.26 / 0.579	20.57 / 0.469	23.38 / 0.589
BSD100	×2	15.52 / 0.440	24.08 / 0.593	23.96 / 0.603	22.35 / 0.530	23.85 / 0.619	24.53 / 0.646	22.53 / 0.536	24.21 / 0.617
	×3	15.58 / 0.435	23.26 / 0.560	22.96 / 0.559	21.21 / 0.484	22.43 / 0.555	23.12 / 0.578	21.85 / 0.461	23.98 / 0.598
	×4	15.33 / 0.427	22.54 / 0.536	22.21 / 0.530	20.71 / 0.462	21.42 / 0.517	22.30 / 0.538	21.02 / 0.449	23.45 / 0.564
Urban100	×2	13.52 / 0.409	22.81 / 0.659	23.36 / 0.713	20.42 / 0.566	22.63 / 0.723	24.18 / 0.753	20.80 / 0.566	24.33 / 0.715
	×3	13.69 / 0.402	21.42 / 0.591	21.55 / 0.627	19.31 / 0.514	20.83 / 0.630	21.01 / 0.661	19.71 / 0.456	21.99 / 0.666
	×4	13.37 / 0.392	20.27 / 0.562	20.47 / 0.569	18.31 / 0.462	19.35 / 0.565	20.01 / 0.580	18.85 / 0.436	21.07 / 0.581
KodaK24	×2	16.31 / 0.544	25.36 / 0.656	25.24 / 0.661	23.51 / 0.592	25.28 / 0.679	26.02 / 0.702	23.56 / 0.578	25.55 / 0.672
	×3	16.35 / 0.540	24.60 / 0.631	24.27 / 0.626	22.74 / 0.562	24.15 / 0.626	24.51 / 0.644	22.97 / 0.511	24.87 / 0.632
	×4	16.10 / 0.534	23.49 / 0.599	23.57 / 0.602	21.80 / 0.541	22.65 / 0.591	23.66 / 0.609	22.17 / 0.508	24.22 / 0.615

TABLE IV
COMPARISON OF AVERAGE PSNR/SSIM PERFORMANCES OF VARIOUS DESPECKLING SR METHODS FOR ×2, ×3, AND ×4 UPSCALING WITH SPECKLES OF FOUR LOOKS (BEST AND SECOND-BEST PERFORMANCES ARE IN RED AND BLUE, RESPECTIVELY)

Datasets	Scale	SRCNN	FSRCNN	CFSRCNN	MemNet	RCAN	RRDBNet	NFGAN	Proposed
Set5	×2	13.40 / 0.471	24.15 / 0.700	25.13 / 0.742	21.76 / 0.604	24.39 / 0.728	25.12 / 0.758	22.50 / 0.634	25.98 / 0.759
	×3	13.48 / 0.462	22.76 / 0.660	23.33 / 0.675	19.79 / 0.532	22.54 / 0.664	23.26 / 0.696	21.30 / 0.541	24.42 / 0.697
	×4	13.07 / 0.447	21.70 / 0.621	22.17 / 0.631	18.83 / 0.491	21.51 / 0.625	22.17 / 0.654	20.18 / 0.533	24.15 / 0.676
Set14	×2	14.60 / 0.444	22.97 / 0.590	23.36 / 0.620	20.55 / 0.502	22.66 / 0.602	23.54 / 0.634	21.30 / 0.518	23.22 / 0.607
	×3	14.62 / 0.438	21.81 / 0.553	22.00 / 0.564	19.44 / 0.461	21.26 / 0.548	22.01 / 0.573	20.37 / 0.448	22.89 / 0.579
	×4	14.30 / 0.427	20.98 / 0.528	20.98 / 0.525	18.89 / 0.451	20.18 / 0.512	20.95 / 0.536	19.46 / 0.437	22.75 / 0.573
BSD100	×2	15.47 / 0.437	22.90 / 0.546	22.87 / 0.555	20.97 / 0.460	22.78 / 0.563	23.23 / 0.587	21.44 / 0.486	23.49 / 0.578
	×3	15.52 / 0.433	22.07 / 0.521	21.91 / 0.518	20.19 / 0.438	21.62 / 0.514	22.00 / 0.530	20.80 / 0.426	23.07 / 0.551
	×4	15.15 / 0.422	21.45 / 0.505	21.19 / 0.496	19.62 / 0.419	20.77 / 0.487	21.26 / 0.500	20.09 / 0.424	22.75 / 0.529
Urban100	×2	13.50 / 0.405	21.24 / 0.586	21.88 / 0.649	19.23 / 0.507	21.61 / 0.659	22.74 / 0.697	19.67 / 0.513	21.95 / 0.658
	×3	13.56 / 0.398	20.17 / 0.537	20.37 / 0.573	17.97 / 0.447	19.87 / 0.574	20.84 / 0.608	18.72 / 0.418	21.19 / 0.592
	×4	13.24 / 0.387	19.41 / 0.504	19.42 / 0.524	17.47 / 0.421	18.65 / 0.517	19.83 / 0.555	17.99 / 0.408	20.41 / 0.543
KodaK24	×2	16.26 / 0.540	24.07 / 0.614	24.11 / 0.621	21.49 / 0.508	24.25 / 0.633	24.72 / 0.653	22.55 / 0.541	24.46 / 0.629
	×3	16.27 / 0.536	23.35 / 0.597	23.16 / 0.591	21.39 / 0.515	22.94 / 0.591	23.37 / 0.605	21.98 / 0.484	24.13 / 0.607
	×4	15.90 / 0.529	22.48 / 0.570	22.54 / 0.573	20.72 / 0.501	22.13 / 0.566	22.58 / 0.579	21.21 / 0.488	23.06 / 0.585

IV. EXPERIMENT

A. Experimental Protocol

1) Speckle Synthesis: We follow the fully developed Goodman model for noise simulation [41]. This model describes SAR speckle as a multiplicative noise, which can be mathematically expressed as follows:

$$Y(i, j) = X(i, j)N(i, j) \tag{8}$$

where Y represents the actual SAR image, X represents the corresponding clean image, N represents the speckle noise, and (i, j) represents the pixel position within the image. For an SAR image with L looks, the speckle noise is assumed to follow the Gamma distribution with a unit mean and $1/L$ variance, formulated by

$$p_N(N) = \frac{L^L N^{L-1} \mathrm{e}^{-LN}}{\Gamma(L)} \tag{9}$$

where $\Gamma(L)$ represents the Gamma function.

Consequently, when provided an image X, we randomly generate Gamma noise using the specific number of looks and apply the noise multiplicatively to the image, resulting in the speckled image $\hat{X}$.

2) Dataset Description: We choose the DIV2K dataset [42] for the training data, which is a widely used dataset for image SR tasks. The DIV2K dataset contains 800 training images, 100 validation images, and 100 test images at three different scales: ×2, ×3, and ×4. To enrich the training

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TABLE V

COMPARISON OF AVERAGE PSNR/SSIM PERFORMANCES OF VARIOUS DESPECKLING SR METHODS FOR ×2, ×3, AND ×4 UPSCALING WITH SPECKLES OF SINGLE LOOK (BEST AND SECOND-BEST PERFORMANCES ARE IN RED AND BLUE, RESPECTIVELY)

Datasets	Scale	SRCNN	FSRCNN	CFSRCNN	MemNet	RCAN	RRDBNet	NFGAN	Proposed
Set5	×2	13.31 / 0.454	20.95 / 0.593	21.94 / 0.634	19.26 / 0.501	21.40 / 0.633	22.10 / 0.668	19.38 / 0.531	23.49 / 0.666
	×3	13.18 / 0.443	20.03 / 0.566	20.59 / 0.583	18.61 / 0.468	19.85 / 0.573	20.50 / 0.605	18.69 / 0.460	23.34 / 0.651
	×4	12.93 / 0.438	19.18 / 0.549	19.64 / 0.550	17.51 / 0.450	18.87 / 0.533	19.39 / 0.563	17.67 / 0.460	22.87 / 0.629
Set14	×2	14.50 / 0.432	20.50 / 0.503	20.90 / 0.532	18.76 / 0.431	20.35 / 0.518	21.09 / 0.546	18.85 / 0.448	22.19 / 0.547
	×3	14.23 / 0.423	19.57 / 0.486	19.72 / 0.491	18.34 / 0.415	19.11 / 0.480	19.75 / 0.500	18.44 / 0.392	21.94 / 0.532
	×4	14.11 / 0.420	19.03 / 0.475	18.93 / 0.466	17.52 / 0.396	18.12 / 0.457	18.87 / 0.477	17.65 / 0.396	21.58 / 0.514
BSD100	×2	15.35 / 0.426	20.85 / 0.479	21.09 / 0.489	19.53 / 0.388	20.86 / 0.491	21.29 / 0.508	19.46 / 0.429	22.29 / 0.512
	×3	15.16 / 0.420	20.18 / 0.471	20.17 / 0.465	19.09 / 0.386	19.84 / 0.463	20.23 / 0.473	19.17 / 0.386	22.12 / 0.500
	×4	14.97 / 0.416	19.59 / 0.464	19.52 / 0.451	18.48 / 0.378	19.11 / 0.447	19.54 / 0.457	18.56 / 0.396	21.90 / 0.486
Urban100	×2	13.42 / 0.392	18.96 / 0.478	19.52 / 0.541	17.38 / 0.413	19.22 / 0.546	20.36 / 0.593	17.62 / 0.428	20.46 / 0.552
	×3	13.21 / 0.384	18.17 / 0.457	18.36 / 0.485	16.91 / 0.387	17.91 / 0.481	18.81 / 0.516	17.08 / 0.360	20.13 / 0.528
	×4	13.08 / 0.380	17.61 / 0.442	17.68 / 0.456	16.35 / 0.367	16.97 / 0.443	17.99 / 0.480	16.52 / 0.366	19.70 / 0.499
KodaK24	×2	16.12 / 0.529	21.81 / 0.547	22.20 / 0.564	20.29 / 0.449	22.20 / 0.572	22.73 / 0.589	20.43 / 0.493	23.32 / 0.565
	×3	15.87 / 0.523	21.23 / 0.550	21.36 / 0.546	20.07 / 0.456	21.07 / 0.546	21.50 / 0.557	20.27 / 0.451	23.27 / 0.554
	×4	15.70 / 0.522	20.63 / 0.533	20.77 / 0.535	19.32 / 0.452	20.22 / 0.533	20.74 / 0.543	19.68 / 0.468	20.96 / 0.546

data, we utilize the entire DIV2K dataset (1000 images) as the training set. The dataset is first converted to grayscale. Then, we obtain the LR versions through bicubic interpolation and divide each LR image into patches of size 192 × 192. To enhance the generalization of model training, we randomly apply augment operations, such as flipping and rotation to patches. Subsequently, multiplicative speckle with a random number of looks ranging from 1 to 20 is added to them when training the student network.

The test data are divided into synthetic and real-world SAR images. For the synthetic data, we select five benchmark datasets, including Set5 [43], Set14 [44], BSD100 [45], Urban100 [46], and KodaK24 [47], which are constructed at three different scales (×2, ×3, and ×4). Set5 and Set14 are small but commonly used datasets containing 5 and 14 high-quality images, respectively. BSD100 is a subset of the Berkeley segmentation dataset, which contains a wide array of natural images with varied textures and complex structures. Urban100 comprises 100 HR images specifically focused on urban scenes. KodaK24 consists of 24 natural images with a variety of scenes, such as landscapes, animals, and human-made structures. These datasets collectively provide a robust framework for testing the joint despeckling and SR methods across different scales and image types, ensuring comprehensive performance evaluation. These datasets also need to be converted to grayscale first. The LR test data are also obtained through bicubic downsampling. Speckle noise of different numbers of looks is added to evaluate the performance of the trained despeckling SR models. For the SAR data evaluation, three SAR images from the website of Sandia National Laboratories [48] are selected, denoted as SAR1, SAR2, and SAR3.

3) Experimental Setup: When FAKD is adopted as the FD loss, the channel numbers of all the MRBs and AMs in the teacher DSRKD$_T$ are set to 128, while the corresponding channel numbers of those modules in the student DSRKD$_S$ are set to 64. When L_1 is adopted as the FD loss, the channel numbers of both DSRKD$_T$ and DSRKD$_S$ are set to 64.

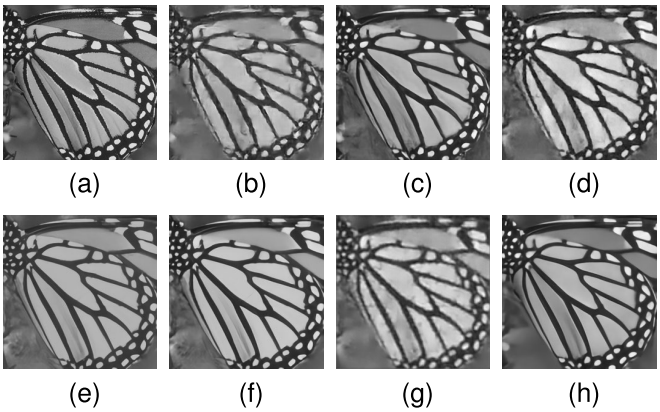


Fig. 5. Despeckled ×2 results of each method for an image from Set5 with $L = 10$ speckle noise. (a) Ground-truth PSNR/SSIM. (b) FSRCNN 26.85/0.778. (c) CFSRCNN 24.08/0.835. (d) MemNet 21.90/0.782. (e) RCAN 24.97/0.861. (f) RRDBNet 24.14/0.852. (g) NFGAN 20.42/0.686. (h) Proposed 25.03/0.857.

During training, a random SR scale is selected from the set $\{2, 3, 4\}$. We use the Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.99$. Both the teacher and student networks are trained from random initialization for 10^5 iterations with a batch size of 4. The initial learning rate is set to 10^{-4} and halved every 4×10^4 iterations. We choose $\lambda_1 = 0.01$ and $\lambda_2 = 0.1$ as the weight factors for the feature and TD loss. In the inference stage, only DSRKD$_S$ will be used. The proposed and all other comparative methods are carried out in PyTorch on the server of NVIDIA A800 GPU.

B. Evaluation Metrics

For synthetic datasets, we employ peak signal-to-noise ratio (PSNR) and SSIM for evaluation. PSNR effectively estimates the fidelity of the image, with a higher value indicating a stronger reconstruction capability of the method. SSIM effectively measures the edge preservation capability of the despeckled image, with a higher value indicating a stronger ability of the method to recover edges. Note that all the

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methods evaluate the performance of PSNR and SSIM on grayscale images.

For real-world SAR data, since the ground truth of SAR images is not available, we select several nonreference metrics for evaluation. The descriptions of these metrics are as follows.

1) *ENL [49]:* The equivalent number of looks is used to measure the relative strength of the speckle noise in the homogeneous area of the image. It can be formulated as $\text{ENL} = \mu_{\hat{X}}^2/\sigma_{\hat{X}}^2$, where $\mu_{\hat{X}}$ and $\sigma_{\hat{X}}$ are the mean and standard deviation of the filtered image. The larger the ENL value, the smoother the homogeneous area, indicating a better filtering effect.
2) *MoR [50]:* The mean of ratio represents the ratio image between the noisy and filtered images $\text{MoR} = Y/\hat{X}$. For an ideal filter, the ratio image should be pure speckle, so MoR should be unitary.
3) *EPD-ROA [51]:* The edge preservation degree based on the ratio of average is used to measure the detail preservation capability of the despeckled image. The closer its value is to 1, the stronger the image detail preservation capability. It can be formulated as follows:

$$\text{EPD-ROA} = \frac{\sum_{i=1}^{m} |I_{\hat{X}1}(i)/I_{\hat{X}2}(i)|}{\sum_{i=1}^{m} |I_{Y1}(i)/I_{Y2}(i)|} \tag{10}$$

where $I_{\hat{X}1}(i)$ and $I_{\hat{X}2}(i)$ represent the adjacent pixel values of the despeckled image along a certain direction (horizontal and vertical directions are selected in our experiments, denoted as HD and VD). Similarly, $I_{Y1}(i)$ and $I_{Y2}(i)$ represent the adjacent pixel values of the speckled image.

C. Ablation Study

To validate the influence of various model components on the performance of DSRKD, we conduct a series of ablation experiments as follows: 1) removing all AMs; 2) removing the skip connections in the encoder; 3) removing the upsampling module corresponding to Z_1; and 4) removing the upsampling modules corresponding to both Z_1 and Z_2 (the upsampling module corresponding to Z_3 must be retained). As illustrated in Fig. 4, the removal of any component leads to a performance decline, demonstrating the effectiveness of our proposed network architecture.

Besides, to verify the effectiveness of our distillation strategy in improving model performance, a series of ablation studies are conducted across different datasets with ten looks. The overall results are presented in Table I. In the first row, only reconstruction loss is adopted, which indicates that the student network is trained without distillation and only supervised by HR images. The subsequent rows represent that the student network is trained using different distillation algorithms. The second and third rows demonstrate that adopting both L_1 and FAKD as FD loss can enhance performance. In addition, only implementing TD is also beneficial, as shown in the fourth row. As depicted in the last two rows, the DSRKD_S achieves the best results when both FD and TD are applied simultaneously, compared with using either distillation method alone. Notably, selecting FAKD for FD yields better results than using the L_1 loss. In the subsequent experiments, we adopt the combination of FAKD for FD and L_1 for TD as our default experimental setting.

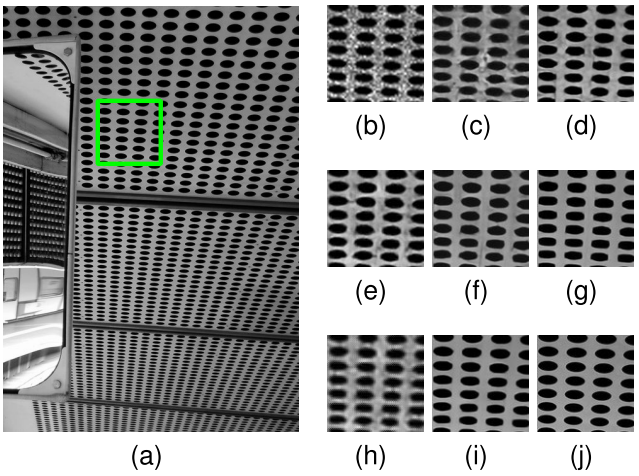


Fig. 6. Despeckled ×2 results of each method for an image from Urban100 with $L = 10$ speckle noise. (a) Original. (b) Bicubic. (c) FSRCNN. (d) CFSRCNN. (e) MemNet. (f) RCAN. (g) RRDBNet. (h) NFGAN. (i) Proposed. (j) Ground truth.

D. Comparison Method

Comparative Methods: To validate the effectiveness of the proposed algorithm, we compared it with state-of-the-art SR algorithms. The comparative algorithms can be categorized as follows.

1) SRCNN [1], FSRCNN [2], CFSRCNN [6], and RCAN [4] are SR algorithms proposed for optical images. We retrain them with paired speckled LR and clean HR images $\{X_{\text{LR}}^n, X_{\text{HR}}\}_{i=1}^N$, where N is the size of the training set.
2) MemNet [3] and Real-ESRGAN [13] are applied to reconstructing from noisy LR optical images, where the noises studied include common optical noises, such as additive Gaussian noise and blocking artifacts caused by JPEG compression. We replace these noises with multiplicative Gamma speckles and retrain these algorithms accordingly. Since we observe that the GAN-based model has difficulty converging when it receives severe speckle noise, especially ESRGAN, which fails to reconstruct images when $L \leq 2$, we utilize only its generator (degenerated to RRDBNet [5]) for training and testing.
3) NFGAN [14] is a GAN-based joint denoising and SR algorithm for SAR images. We reproduce the algorithm according to the authors' introduction in this article.

Note that most compared SR methods either lack inherent denoising capabilities or are trained for denoising requirements in the fields of optical images. Consequently, we meticulously replicate all the algorithms based on the respective papers using the same training set.

The comparison of space and time complexities of different SR methods for ×2 upscaling is illustrated in Table II. The runtime complexity is compared on the butterfly image (Fig. 5) with a size of 256 × 256. Since we adopt a relatively shallow

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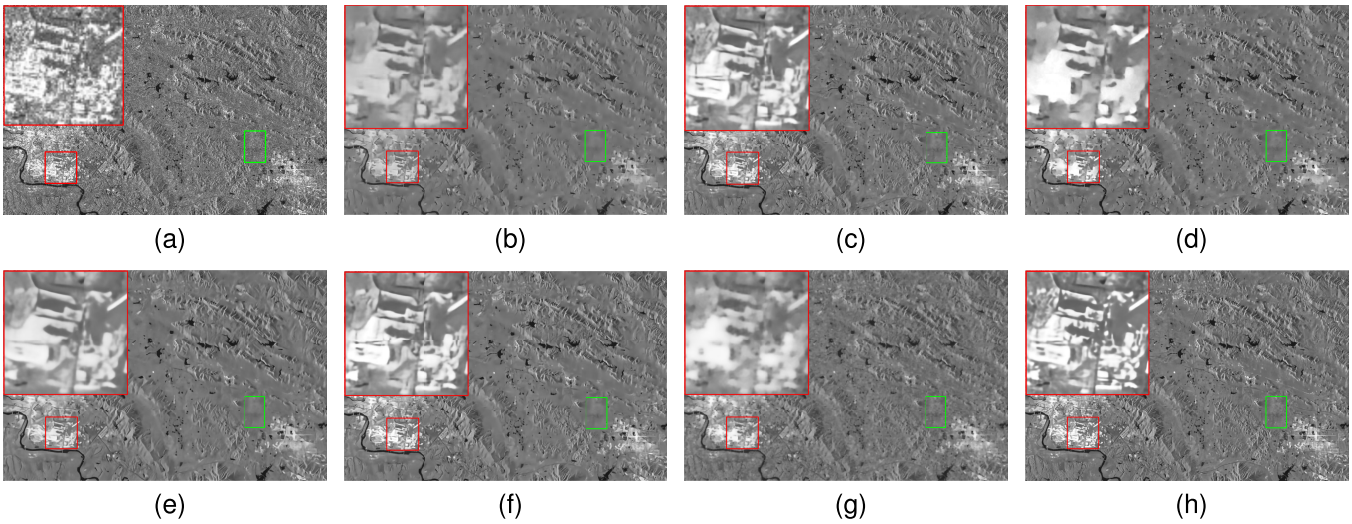


Fig. 7. Despeckled ×4 images of Sentinel-1 by each method. (a) Original. (b) FSRCNN. (c) CFSRCNN. (d) MemNet. (e) RCAN. (f) RRDBNet. (g) NFGAN. (h) Proposed.

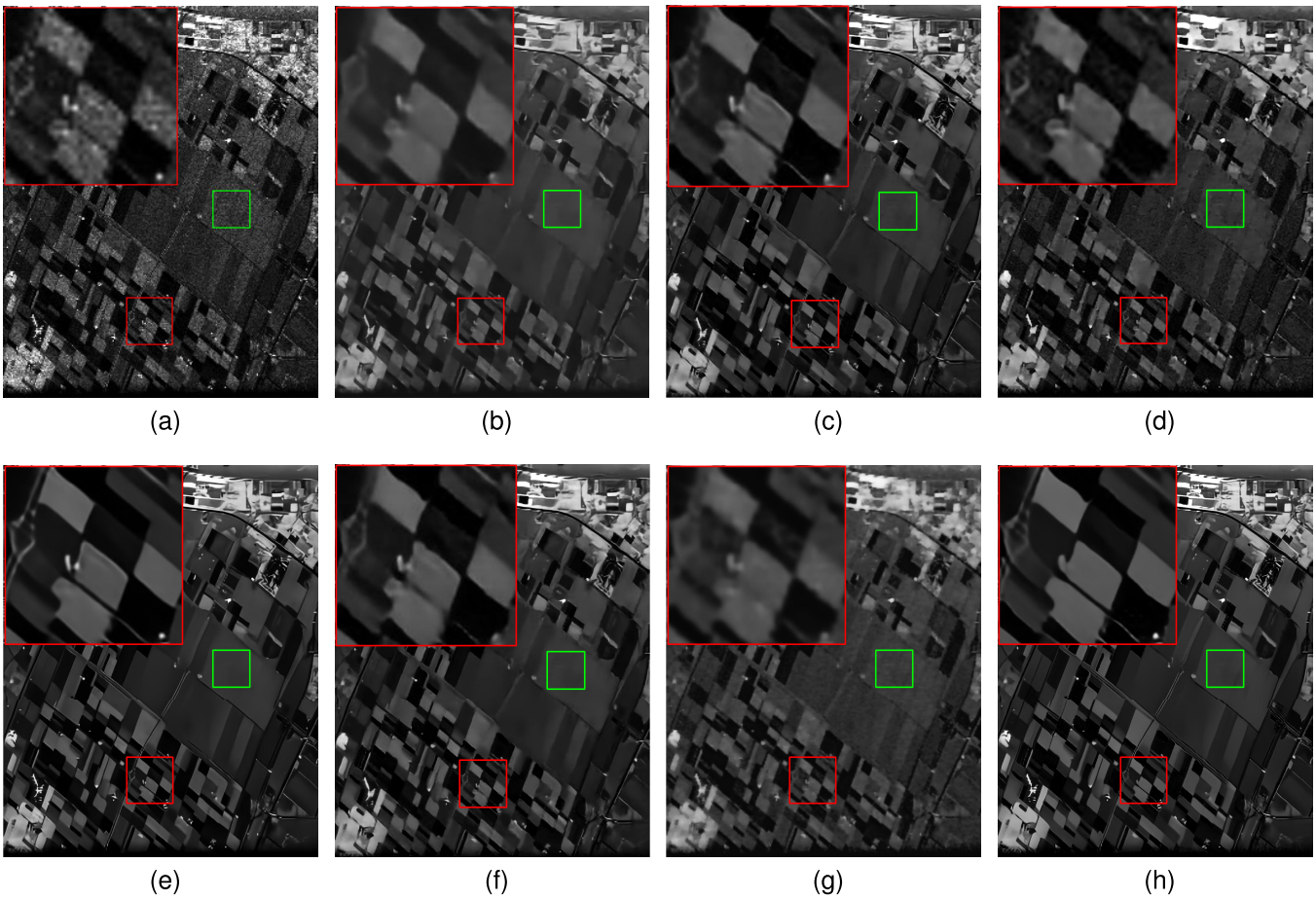


Fig. 8. Despeckled ×4 images of AirSAR by each method. (a) Original. (b) FSRCNN. (c) CFSRCNN. (d) MemNet. (e) RCAN. (f) RRDBNet. (g) NFGAN. (h) Proposed.

architecture and do not use the concatenation operations commonly used in SR, our student model's parameter count is moderate among these algorithms, and the runtime is only higher than that of FSRCNN and CFSRCNN. Nevertheless, owing to the enhancement provided by the distillation strategy, our method achieves the best performance for both visual quality and quantitative metrics in most cases.

E. Simulated Image Despeckling SR Results

For simulated data, we introduce speckles with different numbers of looks across five public datasets and test all the algorithms at different SR scales. The average PSNR and SSIM values obtained by different despeckling SR algorithms on these five datasets are presented in Tables III–V.

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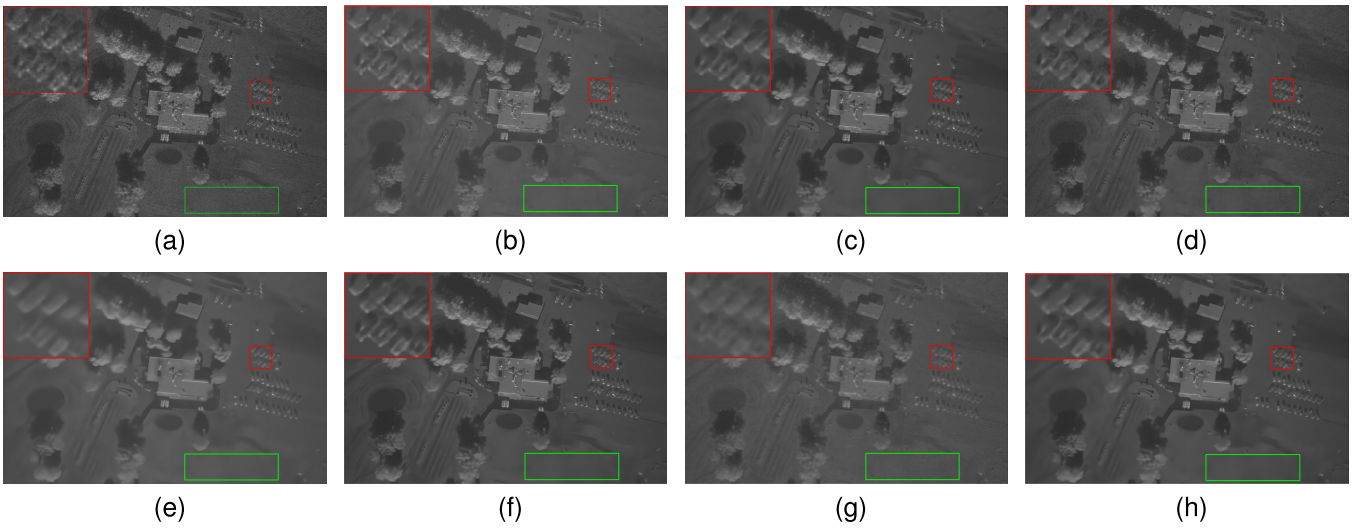


Fig. 9. Despeckled ×4 images of Sandia by each method. (a) Original. (b) FSRCNN. (c) CFSRCNN. (d) MemNet. (e) RCAN. (f) RRDBNet. (g) NFGAN. (h) Proposed.

As illustrated in the tables, our method can achieve the best PSNR and SSIM across most combinations of datasets, speckle levels, and SR scales. At ×2 scale, RRDBNet often achieves better numerical results than our method. However, our algorithm shows more significant advantages as noise intensifies and the SR scale increases. Furthermore, except for the poor denoising performance of SRCNN, other algorithms demonstrate noise reduction capabilities after being trained with paired speckled LR and clean HR images. Notably, FSRCNN achieved very good results in the PSNR metric (second only to our method in many cases), but its SSIM metric is significantly lower than several other algorithms (CFSRCNN, RCAN, RRDBNet, and our DSRKD$_S$). This highlights that a higher PSNR does not necessarily represent a better visual effect in the joint denoising and SR task.

In order to clearly illustrate the difference in visual effects among various denoising methods, we present the global and detailed reconstruction results of the despeckled SR image in Figs. 5 and 6. Fig. 5 illustrates the despeckling effects of each method on the butterfly image. We observe that all the algorithms can despeckle the image when performing SR. FSRCNN achieves the highest PSNR, but its SSIM is the lowest among all methods except NFGAN, and it produces very noticeable artifacts in the reconstructed image. Visually, the results of RCAN, RRDBNet, and our DSRKD$_S$ are the best, with our PSNR and SSIM values being the highest. Based on a comprehensive analysis of objective metrics and visual evaluation, our method can reconstruct despeckled and super-resolved images that are closest to clean images.

In Fig. 6, we enlarge a patch of the despeckled SR images to observe the ability of different algorithms to preserve image texture details. Compared with directly upsampling the noisy image, these methods all achieve some level of despeckling effect, but the issue of artifacts still exists, as shown in Fig. 6(c)–(f). Except for our method, RRDBNet also effectively removes artifacts, but slight distortions are introduced to the ellipses in the image after restoration. Therefore, our proposed method demonstrates the strongest capability in preserving the details of despeckled SR images.

F. SAR Image Despeckling SR Results

To better evaluate the joint denoising and SR performance of different methods, the above seven despeckling SR algorithms (SRCNN's performance is too poor to be included in the comparison) are tested on real-world SAR images. To validate the effectiveness of these algorithms for various SAR images, we select three images from Sentinel-1, AirSAR, and Sandia National Laboratory (referred to as Sentinel-1, AirSAR, and Sandia, respectively). All the SAR images are single-looked. Sentinel-1 is a C-band, VV polarization satellite SAR image with a resolution of 5 and 20 m in the azimuth and range directions, including scenes of mountains and urban areas. AirSAR is an airborne SAR image with an L-band, HH polarization, and a resolution of 6.662 m in range spacing and 12.155 m in azimuth spacing, primarily depicting farmland. Sandia is an HR SAR image with a resolution of 0.1 m (range and azimuth). This image clearly shows details, such as cars, houses, and trees, making it suitable for verifying the algorithm's ability to preserve image details during joint SR and denoising. We perform ×4 SR on all SAR images, and the reconstructed images are shown in Figs. 7–9. The homogeneous area within the green rectangular box is used to calculate ENL.

Visually, these figures demonstrate that all the algorithms can despeckle and super-resolve SAR images to a certain degree. In Fig. 7, we observe that none of the algorithms can effectively handle the urban area, which is challenging due to the high dynamic range and heterogeneity. This finding encourages us to design more realistic SAR models for LR urban scenes in future work. Compared with other algorithms, our method achieves the clearest results. In Fig. 8, we test the filtering capability for homogeneous farmland regions and the edge preservation ability. In the enlarged regions, it is clear that the boundaries between adjacent farmlands reconstructed

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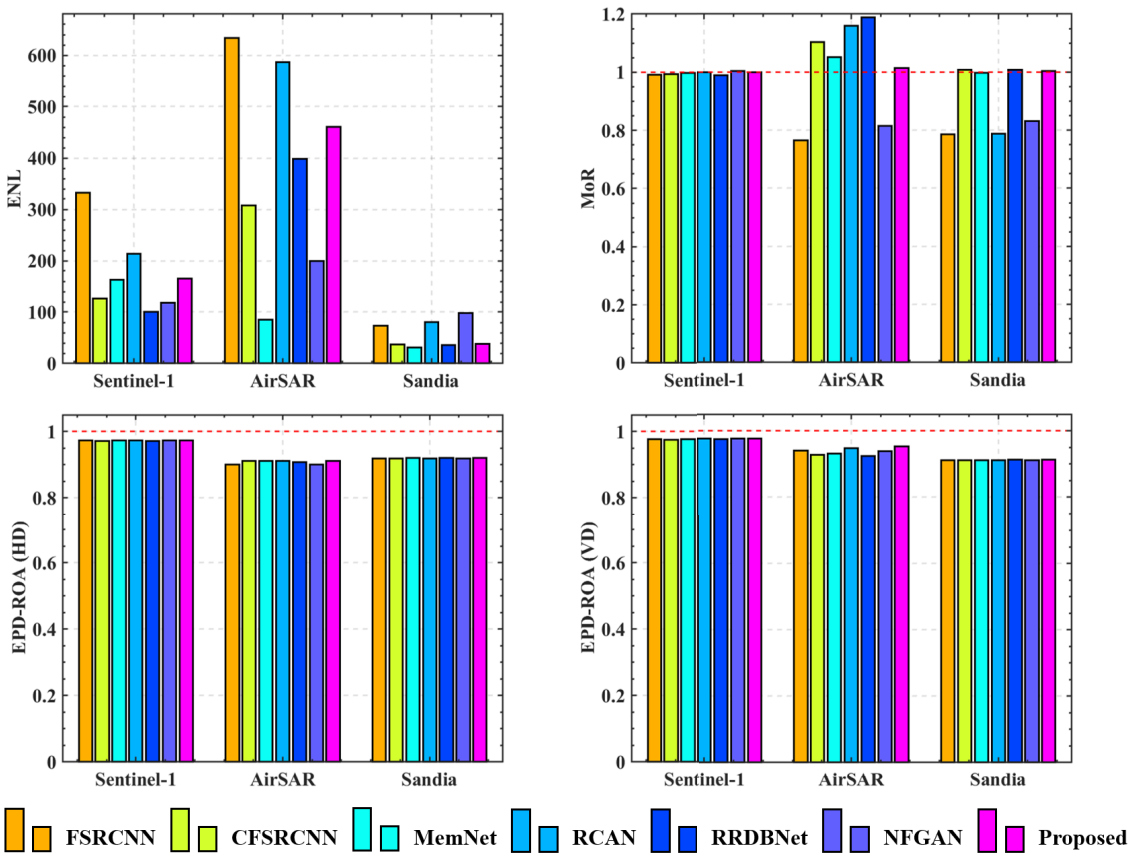


Fig. 10. SAR evaluation metrics on Sentinel-1, AirSAR, and Sandia. Each bar represents the metrics measured by each method. The red dashed line represents the ideal value of MoR or EPD-ROA. The higher the ENL value, the better the reconstruction performance.

by methods, such as FSRCNN, MemNet, RRDBNet, and NFGAN, have become blurred. RCAN achieves good edge reconstruction results, but there are slight artifacts on the farmlands. Our method best reconstructs the despeckled and super-resolved farmlands. However, upon close observation, our reconstructed edges still show slight discrepancies from the original edges due to noise contamination. In Fig. 9, FSRCNN and MemNet exhibit the poorest despeckling capability, as speckles still exist in the reconstructed images. RCAN and NFGAN suffer from excessive smoothing, leading to the loss of many fine texture details present in the original SAR images. RRDBNet and our method show a good balance between denoising and preserving details, reconstructing SAR images that maintain the best visual presentation after being upscaled four times.

As for the quantitative results, the nonreference evaluation metrics for each method are depicted in Fig. 10. It is noted that FSRCNN and RCAN exhibit significant advantages in the ENL metric, indicating their strong smoothing capability compared with other algorithms. However, it is precisely the issue of oversmoothing that results in their MoR metrics being farthest from 1, especially for AirSAR and Sandia, which means that the filtered noise no longer follows the Gamma distribution. Across the three SAR images, our method achieves MoR values closest to 1, indicating that our model effectively learns the distribution of speckle noise and filters it out from the images. Besides, among the algorithms with MoR values differing from 1 by no more than 0.1 on AirSAR (CFSRCNN, MemNet, and DSRKD$_S$) and on Sandia (CFSRCNN, MemNet, RRDBNet, and DSRKD$_S$), our method achieves the highest ENL values. As for the EPD-ROA indicator, the values of this metric for all algorithms are very close. In summary, in SAR scenarios, our method can most effectively learn the distribution of speckles in SAR images and best filter out noise while fully preserving image information.

V. Conclusion

Most existing SR methods do not take into account scenarios where images are severely contaminated by speckle noise. When these methods are directly applied to SAR images, the speckle is also amplified, leading to a significant degradation in image quality. To address these problems, we introduce a novel KD-based joint despeckling and SR method called DSRKD in this article. We propose designing the SR network as an encoder–upscaling–decoder architecture, and by introducing the distillation strategy in the encoding phase, speckle is eliminated in the latent space, thereby restoring SR images free from noise. Through extensive experiments on synthetic data and SAR data, we demonstrate the significant performance improvement of the distillation strategy on the model, and our method outperforms other despeckling SR methods under various scales and noise intensities. We aim to encourage the exploration of multitask joint processing for SAR images, rather than treating denoising merely as an

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upstream processing step. In future research, we will attempt to train our models using real-world multitemporal SAR data and explore more realistic SAR simulation methods. We will employ the distillation mechanism for more joint despeckling tasks with more powerful models.

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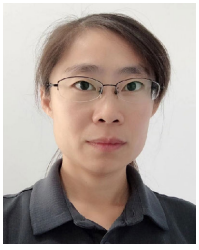
Ziyuan Liu received the B.S. degree from the Department of Electronic Engineering, Tsinghua University, Beijing, China, in 2022, where he is currently pursuing the Ph.D. degree with the Department of Electronic Engineering, Beijing National Research Center for Information Science and Technology.

His research interests include image super-resolution, image denoising, image compression, and SAR image processing.



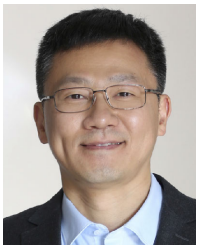
Shaoping Wang received the B.S. degree from the Department of Electronic Engineering, Tsinghua University, Beijing, China, in 2022. He is currently pursuing the M.S. degree with Tsinghua Shenzhen International Graduate School, Shenzhen, China.

His research interests include radar image processing and deep learning.



Ying Li received the B.E. and Ph.D. degrees in communication engineering from the National University of Defense Technology, Changsha, China, in 2001 and 2006, respectively.

She is currently a Researcher with the Peng Cheng Laboratory, Shenzhen, China. Her research interests include network architecture design, MIMO, and cognitive radio.



Yuantao Gu (Senior Member, IEEE) received the B.E. degree in electronic engineering from Xi'an Jiaotong University, Xi'an, China, in 1998, and the Ph.D. degree (Hons.) in electronic engineering from Tsinghua University, Beijing, China, in 2003.

In 2003, he joined the Faculty of Tsinghua University, where he is currently a Professor with the Department of Electronic Engineering. During his career, he also served as a Visiting Scientist at Microsoft Research Asia, Beijing, from 2005 to 2006; at the Research Laboratory of Electronics, Massachusetts Institute of Technology, Cambridge, MA, USA, from 2012 to 2013; and at the Department of Electrical Engineering and Computer Science, University of Michigan, Ann Arbor, MI, USA, in 2015. His research interests encompass high-dimensional statistics, sparse signal recovery, temporal-space and graph signal processing, and related topics in wireless communications and information networks.

Dr. Gu has been an Elected Member of the IEEE Machine Learning for Signal Processing (MLSP) Technical Committee since 2019. He was also an Elected Member of the IEEE Signal Processing Theory and Methods (SPTM) Technical Committee from 2017 to 2022. In recognition of his contributions, he has received several awards, including the Best Paper Award at IEEE GlobalSIP in 2015, the Award for Best Presentation of a Journal Paper at IEEE ChinaSIP in 2015, the Zhang Si-Ying (CCDC) Outstanding Youth Paper Award in 2017 (shared with one of his students), and the Best Paper Award at ICCSPS in 2019. In addition, he has received the Outstanding Reviewer Award and the Outstanding Editorial Board Member Award for his services at IEEE ICASSP in 2019 and for IEEE TRANSACTIONS ON SIGNAL PROCESSING in 2021, respectively. From 2019 to 2023, he served as a Senior Area Editor for IEEE TRANSACTIONS ON SIGNAL PROCESSING. Prior to that, he was an Associate Editor of IEEE TRANSACTIONS ON SIGNAL PROCESSING from 2015 to 2019 and a Handling Editor of *Digital Signal Processing* (Elsevier) from 2015 to 2017.



Quan Yu (Fellow, IEEE) received the B.S. degree in radio physics from Nanjing University, Nanjing, China, in 1986, the M.S. degree in radio wave propagation from Xidian University, Xi'an, China, in 1988, and the Ph.D. degree in fiber optics from the University of Limoges, Limoges, France, in 1992.

In 1992, he joined the faculty of the Institute of China Electronic System Engineering Corporation, Beijing, China, as a Senior Engineer. He is currently a Research Fellow of Peng Cheng Laboratory, Shenzhen, China. His research interests include the architecture of wireless networks, optimization of protocols, and cognitive radios.

Dr. Yu is an Academician of Chinese Academy of Engineering (CAE) and the Funding Editor-in-Chief of *Journal of Communications and Information Networks* (JCIN).

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